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How physio-virtual-digital connections are changing the face of Manufacturing in India

Manufacturing has had a long history of change. From the early days of steam-powered machines to high-volume assembly lines to automation and computer systems, technology has played a vital role in driving innovation. The world of manufacturing has entered the Fourth Industrial Revolution – a time when the Internet has enabled unprecedented levels of communication between machines, products and people. At the same time, a digital revolution is underway. India too is at the cusp of this revolutionary change.

According to the World Economic Forum, India is the 5th largest manufacturer in the world¹ with a total manufacturing value added of over USD 420 billion in 2016. India's manufacturing sector grew by over 7 per cent per year on average during the past three decades accounting for 16-20 percent of the country's GDP. But is past performance an indicator for what the future is going to look like?

While the first industrial revolution was all about mass production, 4.0 is about using technology to move from mass production to mass customized production, and it's already happening at a rapid pace. And this is allowing

manufacturers to react to consumer demand more efficiently.

But let's take a pause and understand what digital transformation does. It brings together every system, device, process, data and asset across the organization using a smart production platform. By harnessing all the data in an integrated way, it enables you to deliver the right expertise, at the right time, to produce higher-quality products more efficiently. This is the vision of the manufacturing Industry 4.0 model powered by digital transformation.

For this, the Internet of Things (IoT) is being harnessed across several use cases. For instance, IoT can provide real-time feedback and alert companies about defects or damaged goods and help reduce cost and waste. Or in the case of IoT sensor data-based preventive maintenance through continuous monitoring, whereby the equipment can inform technicians before services are needed so they can proactively address issues and prevent costly downtime. Operational efficiencies gained through predictive maintenance can help manufacturers transition from being reactionary to having predictive—even pre-emptive—maintenance capabilities, im-

mediately increasing their bottom lines. In some cases, digitizing manufacturing and business processes has even resulted in a 45 percent increase in customer satisfaction.

Another example is manufacturers seizing new sources of revenue by connecting outside of their organizations directly with customers and breaking down the silos between salespeople and service representatives, engineers, production/warehouse workers, and delivery drivers. There is also the rise of smart machinery, which has introduced a hybrid approach of virtual and actual content warehouses freeing up manpower on the production and collaboration side of the industry.

Going a step ahead, digital transformation impacts every aspect of operations and the supply chain. It starts with equipment design and continues through product design, production process improvement, and ultimately, monitoring and improving the end user experience. Cloud computing services are in fact helping manufacturers in building, testing, deploying and managing applications and services through a global network of managed datacenters.

The physical-virtual-digital connection leveraging new cloud-based and IoT technologies such as machine learning, advanced analytics, and artificial intelligence is helping manufacturers connect previously siloed business areas and revolutionize the way they engage with increasing technology-based methods. This advancement is the canvas on which solutions for the future need to be built, to continue evolving to meet the changing needs of customers and capture new business opportunities. Manufacturers need to strategize digital scenarios, develop use cases, and build a smart production platform to increase productivity and grow business.

Industry 4.0 is here to stay. The question is – how quickly can manufacturers in India benefit from it. The World Bank predicts 7.5 percent CAGR for India by 2021. With digital transformation, especially in the manufacturing sector, India can benefit by an additional 1 percent CAGR. That's US\$154 bn of additional GDP³. That's a number for you to think about. **d**